BoardQuest Admin

Bi-Weekly Engineering & Architectural Changes Report

RELEASE & AUDIT REPORT

Scope: `apps/admin`
Timeline: Aug 20, 2026 – Sep 03, 2026 (Until Now)
Generated: September 03, 2026

TOTAL COMMITS	FILES MODIFIED	LINES INSERTED	LINES DELETED
22	194	+16,342	-8,163
14-Day Cycle (Until Now)	Across Core Modules	New Features & Refactors	Net +8,179 Lines

EXECUTIVE SUMMARY

During the two-week development sprint between **August 20, 2026** and **September 3, 2026 (present)**, the `apps/admin` codebase underwent a transformative modernization. The legacy `Faculty` subsystem was completely replaced by an enterprise-grade **Universal User & Role Architecture**. Key innovations include the deployment of a high-speed **Fast Review Workspace** with keyboard hotkeys, rich multimedia support (MCQ choice images and PDF module previewers), a 360-degree **Review Center Details & Calendar Engine**, dynamic **Database Analytics** (purging static mockups), **Pusher WebSockets** real-time event streaming, full **Game Badges CRUD**, and a modernized **Global Modal & Toast System**.

1 User & Identity Modernization (Faculty → Users)

1.1 Complete Architecture Migration: Universal User Factory

MAJOR ARCHITECTURE

Replaced the legacy `features/faculty` subsystem with a unified `features/users` module that handles all organizational roles under a standardized entity model.

- **Unified Multi-Role Hierarchy:** Standardized management for Super Admin, Admin, Teacher, Team Lead, and External Review Center accounts within `UserManagement.tsx`, `AddUser.tsx`, and `ViewUser.tsx`.
- **Bulk User Import & Live Parsed Entity Editor:** Added `BulkImportUserModal.tsx` and `EditParsedUserDialog.tsx`, allowing administrators to bulk-import CSV rosters, inspect parsed candidates, fix validation issues inline, and batch-create accounts.
- **Normalized Identity Flow:** Completely eliminated legacy `username` fields across login, forgot-password, and profile schemas, standardizing exclusively on verified email authentication.
- **Domain Invitation Resend:** Implemented domain-aware invitation resend triggers directly within user creation and profile views.

1.2 Authentication Decoupling: Account Status vs Session Activity

SECURITY & AUTH

Eliminated false 403 authorization lockouts during login sessions:

- **Decoupled Status Fields:** Cleanly separated persistent administrative authorization status (`accountStatus: ACTIVE | DEACTIVATED`) from transient session telemetry (`isActive`, `lastActiveAt`).
- **Robust Auth Provider:** Auth guards and login middleware now evaluate `accountStatus` strictly for access authorization, while telemetry fields track real-time faculty online presence.

2 Question Bank & Fast Review Innovation

2.1 Fast Review Workspace & Automated Moderation Lifecycle

PRODUCTIVITY

Built a rapid-fire review environment for reviewers and curriculum team leads:

- **Keyboard Shortcuts Workspace:** Introduced `FastReviewWorkspace.tsx` and `FastReviewShortcutsModal.tsx` supporting instant hotkeys: Approve (`A`), Reject (`R`), Flag (`F`), Skip (`S`), and Arrow navigation.
- **Integrated Choice Visualizer & Inline Discussion:** Renders multiple-choice alternatives with immediate correctness indicators and contextual inline comment threads.
- **Lifecycle Status Synchronization:** Automatically transitions question states based on comment resolution and notifies authors upon review decisions.
- **Teacher Visibility Isolation:** Refactored backend queries in `useQuestionBank.ts` to cleanly isolate personal drafts from the approved shared curriculum library.

2.2 Rich Multimedia: MCQ Choice Images & PDF Learning Module Engine

RICH MEDIA

- **Choice-Level Diagram Uploads:** Built `questionImageUploadService.ts`, updating `MCQInput.tsx` and `QuestionInput.tsx` to support high-resolution image uploads for question stems and individual answer choices.
- **Subject Curriculum Modules:** Created `SubjectModulesTab.tsx`, `ModuleModals.tsx`, and `moduleUploadService.ts` for managing PDF learning modules per subject.
- **Embedded PDF Previewer:** Added `PdfPreviewModal.tsx` for inline verification of curriculum PDFs without leaving the dashboard.
- **Targeted Approval Modals:** Replaced deprecated bulk revision modals with dedicated dialogs (`ApproveAuthorQuestionsDialog.tsx`, `AuthorReviewFeedbackModal.tsx`).

3 Review Center Operations & Scheduling System

3.1 360-Degree Review Center Details Hub

MANAGEMENT

Engineered a dedicated multi-tab dashboard (`ReviewCenterDetailsPage.tsx`):

- **Overview Tab (`ReviewCenterOverviewTab.tsx`):** Displays operational metrics, capacity, assigned faculty, and center metadata.
- **Candidate Roster (`ReviewCenterStudentsTab.tsx`):** Displays enrolled review candidates with live game progression badges and account statuses.
- **Assessments & Score Matrix (`ReviewCenterAssessmentsTab.tsx`):** Visualizes candidate assessment scores, evaluation breakdowns, and performance trends.
- **Strict School Year Normalization:** Enforced numeric `YYYY-YYYY` input formatting and normalization across student workflows.

3.2 Review Calendar Engine & Scoped Permission Safeguards

CALENDAR

- **Calendar UI & Pagination:** Upgraded `ReviewCalendarView.tsx` with pagination, event clustering, month navigation, and schedule inspection modals.
- **Scoped Permission Matrix:** Restricted administrative schedule modifications to game final reviews while syncing review center schedules in read-only mode for faculty alignment.

4 Analytics, Real-Time Architecture & Gamification

4.1 Dynamic Database Analytics & Purge of Static Presets

DATA ARCHITECTURE

- **Elimination of Mock Constants:** Completely deleted `constants/subjects.ts`, replacing static presets with real-time DB queries.
- **Responsive Analytics Suite:** Overhauled `OverviewTab.tsx`, `PerformanceGraphsTab.tsx`, `SyllabusReadinessTab.tsx`, `AchievementsTab.tsx`, and `ActivityTimelineTab.tsx` for responsive desktop and mobile viewports.
- **Live Dashboard Counters:** Real-time subject statistics, question counts, and passing rate analytics bound directly to API feeds in `DashboardKpiCards.tsx` and `SubjectOverviewSection.tsx`.

4.2 Real-Time Event Infrastructure (Pusher WebSockets & SSE)

REAL-TIME

- **Pusher WebSocket Integration:** Integrated `lib/pusher.ts` and `usePusherNotifications.ts` for instant push updates during review moderation and user status changes.
- **SSE Stream Isolation:** Maintained `useNotificationStream.ts` for reliable per-user read isolation and instant badge incrementing.

4.3 Game Badges CRUD Engine & Student Profiling

GAMIFICATION

- **Badge Management Engine:** Created `useGameManagement.ts`, `BadgeEditorModal.tsx`, `BadgesManagementTab.tsx`, and `badgeUploadService.ts` for full CRUD manipulation of in-game badges.
- **Profile Integration:** Dynamic rendering of unlocked and in-progress badges inside `StudentBadgesSection.tsx` and `ViewStudent.tsx`.

5 Infrastructure, UI Framework & Build Polish

5.1 Decoupled Modal Store & Modern Toast Engine

FRAMEWORK

- **Global Modal System:** Introduced `modalStore.ts`, `GlobalModalContainer.tsx`, `CustomDialogRenderer.tsx`, `useModal.ts`, and `useConfirm.ts` for clean, promise-based modal dialog orchestration.
- **Enhanced Toast Alerts:** Modernized `utils/toast.tsx` with smooth animations, progress feedback, and contextual toast alerts.
- **Build & Tooling Fixes:** Resolved API DLL file-locking, fixed strict TypeScript compilation issues, tuned Vite configuration, and optimized Vercel build scripts.

6 Complete 22-Commit Chronological Changelog

COMMIT	DATE	TYPE	COMMIT SUMMARY & SCOPE
`1ce4c16`	2026-09-01	FEAT	Update calendar UI, add pagination, polish score management & review assessments tab
`ffec2ba`	2026-09-01	FEAT	Refine review center schedule permissions, school year formatting, and candidate roster
`22f8476`	2026-08-31	FEAT	Added review center calendar feature and school year (sy) integration
`8558cda`	2026-08-30	FEAT	User creation form enhancements, domain-based resend invitation flow
`ca32e4c`	2026-08-29	FEAT	Integrate Pusher WebSockets real-time notification engine into layout & widget
`9c1acdc`	2026-08-29	FIX	Update Vercel build script command configuration for admin package
`a7cb3fb`	2026-08-29	FIX	Resolve API DLL file-locking and fix admin TypeScript compilation errors
`71c0f75`	2026-08-29	REFACTOR	Responsive analytics, enforce DB-driven subjects, build Review Center details suite
`8cb425b`	2026-08-28	FEAT	Fast Review mode with hotkeys, teacher visibility fix, live DB dashboard counts
`4f79c16`	2026-08-27	REFACTOR	Migrate Faculty → Users management architecture, user factory, toast system
`ccbe331`	2026-08-25	FEAT	Decouple accountStatus from activity status, add PDF preview modal, global modal store
`09940d2`	2026-08-23	FEAT	Question stem & MCQ choice image uploads, navigation label refinements
`2a235d2`	2026-08-22	FEAT	Separate Review Center account creation with EXTERNAL role, standardized filters
`74b21c1`	2026-08-21	CHORE	Polished badge visualizations in student view profile page
`d38d94f`	2026-08-21	FEAT	Full Game Badges CRUD implementation and asset upload management
`6f1bead`	2026-08-21	CHORE	Codebase lint cleanup, remove legacy lint output, harmonize role flags
`6c548ec`	2026-08-21	DOCS	Updated Admin module documentation and architecture guide
`c503736`	2026-08-21	DOCS	Capstone README update and setup guide alignment
`2d828c2`	2026-08-21	CHORE	Removed legacy username instances in favor of email identity across admin forms
`0e530f0`	2026-08-21	CHORE	Added author approval dialog for question review feedback
`717602d`	2026-08-21	CHORE	Cleaned up redundant organization filters on Review Center management page
`5c8e46b`	2026-08-21	CHORE	UI label adjustments, badge upload service integration, and localization updates

7 Module Impact & Distribution Matrix

MODULE / DOMAIN	KEY MODIFIED FILES & COMPONENTS	FUNCTIONAL IMPACT
Users & Access	`AddUser.tsx`, `ViewUser.tsx`, `UserManagement.tsx`, `BulkImportUserModal.tsx`, `EditParsedUserDialog.tsx`, `useUsers.ts`	Replaced Faculty with Universal User Factory, bulk import with inline editor, and role permissions.
Question Bank	`FastReviewWorkspace.tsx`, `MCQInput.tsx`, `PdfPreviewModal.tsx`, `SubjectModulesTab.tsx`, `useQuestionBank.ts`	Hotkeyed Fast Review, question/choice image uploads, PDF syllabus viewer, and shared library isolation.
Review Center	`ReviewCenterDetailsPage.tsx`, `ReviewCenterStudentsTab.tsx`, `ReviewCalendarView.tsx`, `SetReviewScheduleModal.tsx`	360° Review Center hub, schedule permission safeguards, student roster, YYYY-YYYY format normalization.
Analytics & KPIs	`AnalyticsPage.tsx`, `OverviewTab.tsx`, `PerformanceGraphsTab.tsx`, `DashboardKpiCards.tsx`, `LeaderboardPage.tsx`	Purged static mockups (`constants/subjects.ts`), dynamic DB counts, and responsive chart layouts.
Game Badges	`useGameManagement.ts`, `BadgeEditorModal.tsx`, `BadgesManagementTab.tsx`, `badgeUtils.ts`	Full CRUD badge management, image asset uploads, and student profile badge synchronization.
Core UI & Infra	`modalStore.ts`, `GlobalModalContainer.tsx`, `toast.tsx`, `usePusherNotifications.ts`, `lib/pusher.ts`	Decoupled global modal store, rich animated toasts, Pusher WebSockets, and build fixes.

8 Architectural Highlights & Verification Summary

8.1 Security & Role Guard Hardening

SECURITY

All sensitive routes (User creation, Game configuration, Review Center management, and Subject Module uploads) enforce multi-tier role verification via `RequireAuth.tsx` and `useRoleFlags.ts`. The separation of `accountStatus` prevents edge-case deactivations while maintaining strict boundary isolation between Admin, Teacher, Team Lead, and Review Center entities.

8.2 Production Build Stability & Test Coverage

RELIABILITY

Build processes across Vite and Vercel have been verified with zero TypeScript compilation errors. DLL file locking in the API bridge was remediated, and unused legacy dependencies were pruned to ensure fast build times and optimal client bundle sizes.

8.3 Environment Configuration & Real-Time Sync

DEPLOYMENT

Deployment requires configuring the updated Pusher environment variables (`VITE_PUSHER_KEY`, `VITE_PUSHER_CLUSTER`) documented in `.env.example` to activate live event channels alongside the SSE fallbacks.